
CARScenes: An Annotation-Only Dataset and Benchmark for Structured Scene Understanding and Failure Analysis in Autonomous Driving

Yuankai He

Wayne State University
yuankai.he@wayne.edu

Weisong Shi

Wayne State University
weisong@wayne.edu

Abstract

CARScenes is an annotation-only dataset and benchmark for structured scene understanding and failure analysis in autonomous driving. We canonicalize 5,192 front-camera frames from Argoverse1, Cityscapes, KITTI, and nuScenes into a versioned JSONL release with one schema-aligned record per frame. The release ships a machine-readable schema, validation and evaluation scripts, fixed split manifests, scenario-slice definitions, and Croissant metadata, while leaving upstream imagery under the original dataset licenses. The benchmark core covers environment, road geometry, traffic control, vehicles, pedestrians, visibility, ego context, and an ordinal severity label intended for stratified analysis rather than policy evaluation. Initial labels were produced with a VLM-assisted pipeline, then reviewed and corrected by a human annotator for every released record. CARScenes v1 freezes 4,592 training records, 500 silver development records for model selection, and a source-balanced 100-frame gold test split with a 25-frame agreement subset. The frozen release already exposes 1,586 traffic-light scenes, 1,781 scenes with vulnerable road users, and utility slices ranging from 28 rainy night-or-dusk scenes to 71 traffic-light scenes in the highest severity bucket. The current manuscript focuses on the reproducible release, the fully human-corrected benchmark protocol, and the scenario slices that make CARScenes useful for failure analysis across legacy driving datasets.

1 Introduction

Autonomous-driving research already benefits from large image and multimodal corpora such as KITTI [10], Cityscapes [6], Argoverse [4], nuScenes [3], Waymo Open [19], and BDD100K [20]. These datasets are foundational for detection, segmentation, tracking, and motion forecasting, but they provide limited support for consistent, per-image scene semantics that can be queried across sources. This gap becomes especially visible when evaluating vision-language models (VLMs): object-centric labels and coarse metadata do not reveal where a model fails under specific contextual conditions such as degraded visibility, visible traffic control, or vulnerable road users at complex intersections.

Recent work on driving VLMs has focused on free-form question answering, instruction following, or direct scenario reasoning [7, 8, 16–18]. Those efforts show that multimodal models can describe traffic scenes, but they do not provide a fixed, schema-aligned benchmark that makes scene-level failures easy to slice, compare, and reproduce across legacy autonomous-driving datasets. As a result, researchers who want to audit model behavior across weather, visibility, traffic control, or road geometry still need to assemble ad hoc metadata or free-text prompts on top of upstream datasets.

Dataset (year)	Modality	Language supervision	Scale	Provenance	Release type
CARScenes (our work)	Single RGB frames	Structured scene KB (28 keys; Severity)	5,192 frames 4,592/500/100 train/dev/audit-cand.	Argoverse1, Cityscapes, KITTI, nuScenes	Annotation-only JSONL schema+validator+splits
Talk2Car [7] (2019)	Front camera (nuScenes)	Commands with bbox (grounding)	8,349/1,163/2,447 cmds	nuScenes	Text + boxes on nuScenes
nuScenes-QA [16] (AAAI'24)	Multi-modal, multi-frame	Programmatic VQA pairs	~460K QAs over ~34K scenes	nuScenes	Benchmark code + data
NuInstruct [8] (CVPR'24)	Multi-view video	Instruction-response (17 subtasks)	91K video-QA pairs	nuScenes (multi-view)	Annotations + loaders
DriveLM-Data [18] (ECCV'24)	Multi-view images/video	Graph-structured QA (P3: perc/pred/plan)	n/a	nuScenes + CARLA	Dataset + eval server
BDD-X [12] (2018)	Video	Action descriptions + explanations	6,970 videos; 77 h ~26K activities	BDD	Explanations dataset

Table 1: Related driving VLM datasets compared to **CARScenes**.

CARScenes addresses this problem as a data-centric dataset and benchmark rather than a control or policy benchmark. We package an annotation-only, schema-aligned release of 5,192 front-camera frames drawn from Argoverse1, Cityscapes, KITTI, and nuScenes. Each record describes scene type, time of day, weather, road conditions, lane structure, traffic signs and lights, surrounding vehicles, pedestrians, visibility, ego context, camera condition, and a coarse severity score. The release is canonicalized into a single JSONL format, validated against a machine-readable schema, and distributed with fixed split manifests, slice definitions, and executable benchmark tooling. This framing is deliberate: CARScenes is meant to support structured scene understanding and failure analysis for driving VLMs, not to stand in for a vision-language-action dataset.

The project repository, including code and release artifacts, is publicly available at <https://github.com/Croquembouche/CARScenes>.

The practical use case is failure slicing. CARScenes makes it possible to ask whether two models behave differently on scenes with adverse visibility and parked vehicles, on intersections containing vulnerable road users, or on traffic-light scenes in higher semantic-difficulty buckets. These are the units of analysis that matter for dataset auditing, scenario mining, and cross-dataset evaluation, yet they are not available in a consistent form from the source datasets themselves.

This paper makes three contributions. First, it turns four legacy driving datasets into a single validated annotation-only release with a stable schema, benchmark splits, and reusable tooling. Second, it documents a VLM-assisted but fully human-corrected annotation workflow, so the released benchmark records should be interpreted as human-reviewed ground truth rather than raw model output. Third, it positions CARScenes as infrastructure for scenario-focused evaluation and data-centric failure analysis, which is the practical contribution highlighted by the IV reviews and the most defensible fit for the NeurIPS Datasets and Benchmarks track.

2 Related Work

Autonomous-driving datasets. KITTI, Cityscapes, BDD100K, Argoverse, nuScenes, Lyft Level 5, and Waymo Open provide multimodal data with object/pixel-aligned labels for detection, tracking, and mapping. These corpora include limited, coarse scene metadata and are not designed for schema-consistent scene-level semantics. CARScenes complements them by releasing annotation-only JSONL scene annotations, schema files, and fixed split manifests for interpretable analysis rather than raw-image redistribution.

Vision-language for driving. BDD-X, Talk2Car, DriveLM/DriveLM-Data, NuInstruct, nuScenes-QA, and recent traffic-scene VLM studies such as Rivera *et al.* [17] pair scenes with free-form text, QA, or direct model analysis. While powerful, supervision and vocabularies are heterogeneous. CARScenes instead offers a versioned, fixed schema enabling consistent querying, benchmark slicing, and cross-dataset comparison.

Scene graphs & risk corpora. Visual Genome [13], RefCOCO [5], and GQA [11] probe general-image relations/referring or VQA [2], while Scenic [9], RiskBench [14], DReyeVE [1], and Deep-Scenario [15] center on simulators, trajectories, or attention. None of these resources provides a

per-image, driving-specific semantic schema grounded directly in real images and packaged as an annotation-only benchmark with fixed scene slices. CARScenes addresses this gap by combining schema-aligned annotations with a reproducible release and evaluation interface.

3 Dataset Overview

3.1 Sources, Frame Selection, and Scope

CARScenes augments four existing autonomous-driving datasets with schema-aligned, scene-level annotations. The current canonical release contains **5,192** front-camera frames from **Argoverse1 (834)**, **Cityscapes (595)**, **KITTI (749)**, and **nuScenes (3,014)**. We release annotations, schema files, split manifests, and benchmark utilities only; original imagery remains governed by the licenses of the source datasets.

Version 1 includes every front-camera frame for which the annotation pipeline produced a parseable label and the release builder could align image and label basenames. We therefore do not claim a corpus-level balancing strategy over scene types or times of day. Instead, we preserve the heterogeneity of the source datasets and introduce balance only when constructing the public development and audit splits.

3.2 Knowledge Base and Benchmark Core

CARScenes was designed for structured scene understanding rather than free-form captioning. The benchmark core keeps the fields that are already used throughout the paper and that can be validated consistently across sources. These fields cover environment, road geometry, traffic control, background-vehicle behavior, ego-vehicle behavior, vulnerable road users, sensor condition, and a discrete Severity label. Auxiliary keys found in the raw annotations (e.g., bicyclists, animals, and policemen) are preserved as side information in the canonical release but are not part of the headline benchmark until they are fully normalized and audited.

3.3 Schema Derivation

The schema was derived from three constraints. First, the fields had to support scenario mining across heterogeneous source datasets rather than within a single benchmark. Second, they had to align with scene descriptors that matter in driving-language work and scenario specification, including infrastructure, road geometry, visibility, agent composition, and ego context [9, 14, 16–18]. Third, they had to be inferable, at least approximately, from a single front-camera frame. This third constraint intentionally excludes action targets, control outputs, and long-horizon behavioral intent; CARScenes is a scene-understanding benchmark, not a driving-policy benchmark.

3.4 Benchmark Fields

3.5 Severity as a Scene-Level Difficulty Signal

CARScenes attaches a bounded integer Severity in the range $[1, 10]$ to each frame. We use severity as a coarse scene-level difficulty signal rather than as a collision-risk or control label. The score reflects visible factors that plausibly make scene understanding harder or more consequential, including scene complexity, traffic-control ambiguity, visibility degradation, crowdedness, and the presence of vulnerable road users or anomalies. The design goal is interpretability: severity is intended for stratified evaluation and failure slicing, not as a surrogate for downstream driving policy quality.

3.6 Canonical Record Format

The canonical release stores one record per frame in JSONL. Metadata fields such as `_id`, `_source`, and `_image_relp` are prefixed with underscores so that benchmark labels remain easy to read and programmatically stable. The machine-readable schema included in the release defines all scalar and list-valued fields used by the evaluator.

Group	Fields (benchmark core)
Environment	Scene TimeOfDay Weather RoadConditions
Road Geometry & Infrastructure	LaneInformation.NumberOfLanes LaneInformation.LaneMarkings LaneInformation.SpecialLanes Directionality
Traffic Control	TrafficSigns.TrafficSignsTypes TrafficSigns.TrafficSignsVisibility TrafficSigns.TrafficLightState
Background Vehicles	Vehicles.TotalNumber Vehicles.VehicleTypes Vehicles.InMotion Vehicles.States
Ego Context	Ego-Vehicle.Direction Ego-Vehicle.Maneuver
Vulnerable Road Users	Pedestrians
Visibility & Sensor	Visibility.General Visibility.SpecificImpairments CameraCondition Severity

Table 2: Benchmark fields retained in the canonical CARScenes v1 release.

3.7 Annotation Workflow and Human Correction

CARScenes uses a VLM-assisted annotation workflow only as a first pass. Initial labels are generated automatically, after which a human annotator reviews every record field-by-field and corrects any wrong labels before release. The benchmark should therefore be interpreted as a set of fully human-corrected annotations rather than as raw model predictions. This distinction matters for the NeurIPS Datasets and Benchmarks track: the contribution is not a VLM labeling demo, but a released benchmark whose final records were manually checked and corrected.

3.8 Annotation and Canonicalization

The released benchmark is defined by the canonicalization pass rather than by the raw JSON dumps. Canonicalization happens after human correction and is intended to normalize schema drift without changing the intended semantic content of the annotations. The release builder normalizes enum drift, moves misplaced fields into their benchmark locations, removes wrapper files, logs unmatched image/label pairs, and emits a stable schema for the resulting JSONL file. This step is essential because the raw annotations contain legacy field variants and auxiliary keys that are useful for auditing but unsuitable for a headline benchmark without cleanup.

4 Release and Benchmark Protocol

4.1 Annotation-Only Release

CARScenes v1 is released as an annotation-only benchmark. The release directory contains the canonical JSONL file, a machine-readable schema, split manifests, benchmark subset files, benchmark slice definitions, Croissant metadata, and validation/evaluation scripts. This choice avoids re-licensing or re-hosting upstream images while still making the benchmark reproducible for researchers who already have access to the source datasets. For a NeurIPS Datasets and Benchmarks submission, the relevant contribution is therefore the annotation layer, schema, and executable benchmark interface, not the original images.

The public repository for the project is <https://github.com/Croquebouché/CARScenes>. This is the reviewer-facing location for the codebase and release materials referenced throughout the paper.

Table 3: Fixed utility slices shipped with CARScenes v1. Counts are measured on the full canonical release.

Slice	Query intent	Count
Night/dusk + rain	adverse lighting and weather	28
VRU + intersection	interaction-heavy urban scenes	115
Traffic light + high severity	controlled but difficult scenes	71
Poor visibility + parked vehicles	clutter under degraded observation	1 323
Roundabout + multiple vehicles	rare geometric interaction case	12

144 4.2 Fixed Splits

145 We freeze three public split roles in the current release:

- 146 • **Train (4,592 records)**: used for model fitting.
- 147 • **Silver-dev-500 (500 records)**: used for model selection and pilot calibration only.
- 148 • **Gold-test-100 (100 records)**: a fixed, balanced held-out split for headline evaluation.

149 The silver development split is balanced across the four sources (**125** records per source). The gold test
 150 split is also source-balanced (**25** records per source) and was selected to emphasize severity diversity,
 151 adverse conditions, vulnerable road users, and traffic-light scenes. We additionally reserve a **25**-record
 152 agreement subset inside the gold test split for secondary spot-checking and future inter-annotator
 153 studies.

154 4.3 Benchmark Tasks

155 The benchmark covers scalar categorical fields, list-valued fields, and the discrete `Severity` label.
 156 Scalar fields are scored with exact-match accuracy. List-valued fields are scored with micro-averaged
 157 precision, recall, and F1. Severity is treated both as an exact-match classification target and as an
 158 ordinal score via MAE, RMSE, and quadratic weighted kappa. We also define fixed scenario slices
 159 for data-centric analysis, including night-or-dusk with rain, vulnerable road users at intersections,
 160 traffic lights under higher severity, poor visibility with parked vehicles, and roundabout scenes with
 161 multiple vehicles.

162 4.4 Utility Slices

163 These slices are the practical reason to use CARScenes. They are not recoverable from a single source
 164 metadata field, and they make it possible to report whether a model fails in particular contextual
 165 situations instead of only reporting an overall score.

166 4.5 Current Benchmark Status

167 The present repository implements the benchmark interface and freezes human-corrected split
 168 manifests. The released records are not provisional: a human annotator reviewed and corrected the
 169 labels before release, and canonicalization then enforced schema consistency on those corrected
 170 records. Accordingly, `silver-dev-500` is reserved for model selection, while `gold-test-100` is
 171 the held-out split intended for headline evaluation.

172 4.6 Why This Benchmark Matters

173 The point of CARScenes is not just to attach more labels to existing images. The release is useful
 174 because it makes failure slicing explicit and reproducible. A researcher can evaluate one model or
 175 many models on the same scene schema, then ask where those models fail under adverse visibility,
 176 around traffic control, in crowded scenes, or under higher semantic difficulty. This is the practical unit
 177 of analysis that is missing from the original source datasets and from most free-form driving-VLM
 178 evaluations.

Table 4: Canonical CARScenes v1 split sizes by source.

Source	Train	Silver-dev	Audit cand.
Argoverse1	684	125	25
Cityscapes	445	125	25
KITTI	599	125	25
nuScenes	2 864	125	25
Total	4 592	500	100

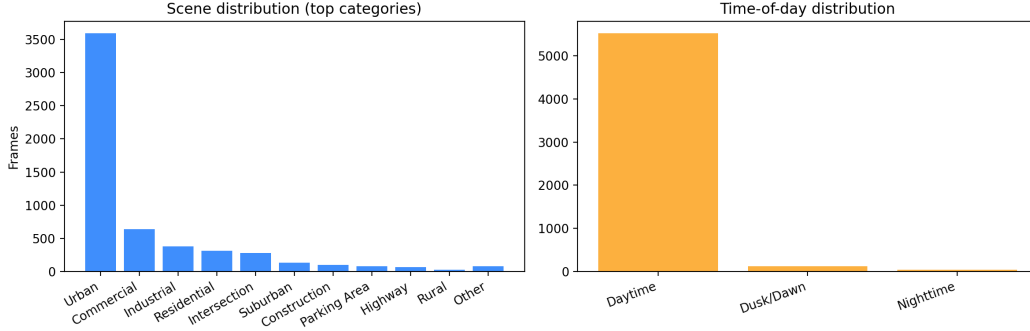


Figure 1: Dataset attribute distribution for representative benchmark fields, including weather, traffic-light state, and vehicle-count marginals.

5 Dataset Statistics and Release QA

This section summarizes the canonical CARScenes v1 release rather than the raw annotation dump. Every count reported here is derived from the generated release artifacts in `release/carscenes-v1/`.

5.1 Source Composition

Figure 1 summarizes the marginal distributions of common scalar fields. Figure 2 visualizes the severity histogram over the full canonical release. The release contains **3,120** low-severity records (1–3), **1,914** mid-severity records (4–6), and **158** high-severity records (7–10). The median severity is **3**, with interquartile range **[2, 4]**.

5.2 Scenario Coverage

The full release contains **5,032** daytime scenes, **111** dusk/dawn scenes, and **39** nighttime scenes. It also contains **1,586** scenes with visible traffic lights, **1,781** scenes with vulnerable road users, and **2,552** scenes tagged as adverse-condition cases by the benchmark slice helper fields. These counts clarify two points raised in the IV reviews: the corpus is intentionally heterogeneous but not balanced, and meaningful scenario mining requires reporting support counts alongside accuracy or F1.

5.3 Release QA

The release builder performs three types of checks before emitting the canonical JSONL file:

- **Record alignment:** image basenames and label basenames must match within each source dataset.
- **Schema normalization:** enum drift, misplaced fields, and optional legacy keys are normalized into the benchmark schema.
- **Split validation:** every split manifest must point to existing canonical records and no train/dev/audit overlap is allowed.

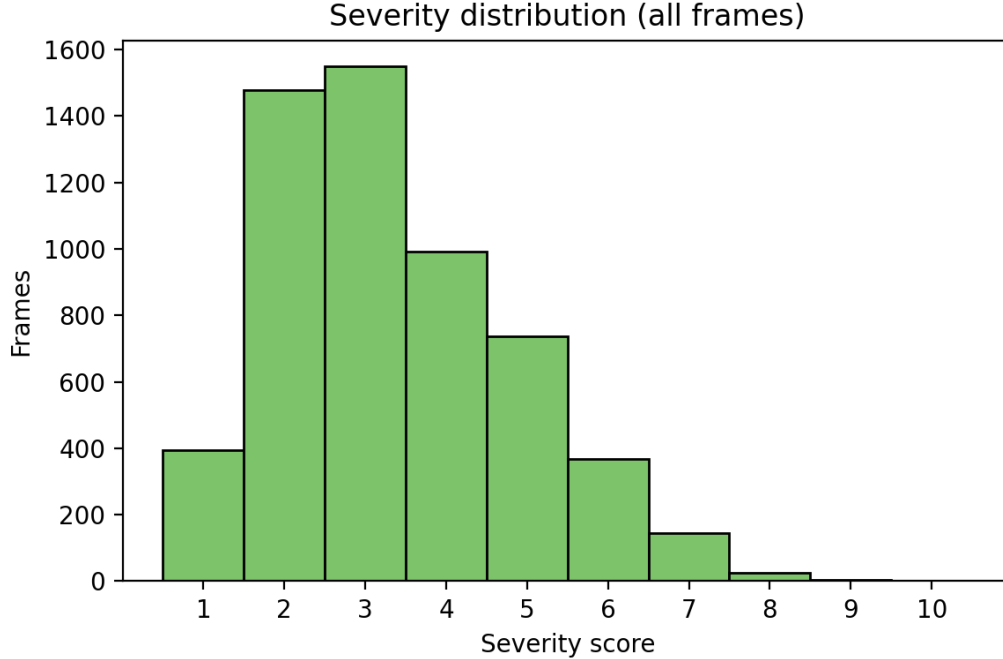


Figure 2: Severity histogram over the full canonical release.

202 In the current snapshot, the canonicalizer recovers **5,192** usable records, skips **3** images without
 203 matching raw labels, and ignores **2** extra raw labels without matching images. The release logger
 204 also records normalization actions for recurrent drift patterns such as misplaced vehicle-type labels
 205 inside the traffic-sign object and spelling or formatting variants in categorical fields. These structural
 206 normalization steps operate on records that have already been manually reviewed and corrected; they
 207 should not be interpreted as replacing human annotation with automatic post-processing.

208 5.4 Coverage Notes

209 CARScenes is intentionally heterogeneous but not perfectly balanced. Daytime scenes dominate the
 210 release, high-severity scenes are rare, and the four source datasets contribute different mixtures of
 211 road geometry and traffic control. We therefore recommend reporting both overall metrics and slice
 212 metrics. In particular, traffic-light state should only be scored where the field is present, and slices
 213 that depend on higher severity or adverse visibility should be reported with support counts alongside
 214 accuracy or F1.

215 5.5 Limitations

216 CARScenes remains a single-frame benchmark. Temporal ambiguity affects some fields more than
 217 others, especially `Ego-Vehicle.Direction`, `Ego-Vehicle.Maneuver`, `Vehicles.States`, and
 218 event-level `Severity` judgments. Interactive demos built on top of these labels should therefore be
 219 treated as illustrative rather than authoritative; without human correction, a VLM can hallucinate
 220 lanes, actor states, or traffic control. The released corpus is fully human-corrected, but the current
 221 paper does not report corpus-wide inter-annotator agreement because the full correction pass was
 222 performed by a primary annotator rather than by multiple independent annotators. Finally, the
 223 benchmark evaluates scene understanding and failure slicing; it does not evaluate a downstream
 224 driving policy.

5.6 Licensing

CARScenes combines our own annotations, schema, and benchmark code with upstream image identifiers from Cityscapes, KITTI, Argoverse1, and nuScenes. Our release artifacts are annotation-only and do not re-license the original images. Users must obtain the source images under the original dataset terms, cite the upstream datasets alongside CARScenes, and treat the original corpora as dependencies rather than as newly released data in this submission.

6 Scenario Utility and Benchmark Status

The central question for a NeurIPS Datasets and Benchmarks paper is whether the resource supports a real scientific use case rather than merely adding more labels. For CARScenes, that use case is slice-aware evaluation. The frozen schema and evaluator let a researcher compare models on the same canonical scene representation, then ask whether failures concentrate around visibility loss, traffic control, dense traffic, vulnerable road users, or higher semantic difficulty.

This contribution is already concrete in the repository. The release includes a deterministic evaluator, fixed split manifests, and public slice definitions. Scalar fields are scored with exact-match accuracy, list-valued fields with micro-averaged precision/recall/F1, and `Severity` with exact-match accuracy, MAE, RMSE, and quadratic weighted kappa. The slice table in Table 3 is therefore not merely descriptive metadata; it defines executable evaluation subsets that can be reused across models and papers.

The current repository state also makes an important positive claim explicit: the held-out split is built from fully human-corrected labels rather than from unaudited model output. This revision therefore removes the earlier provisional framing from the IV draft and treats `gold-test-100` as a valid held-out benchmark split. The stronger scientific position is to separate label validity, which is provided by the human correction pass, from benchmark breadth, which can still be expanded with additional baselines and slice analyses.

With the human correction pass complete, the benchmark should now be able to answer three questions cleanly. Which fields in structured scene understanding are already easy for modern VLMs and which remain weak? How much do overall scores hide differences across source datasets and scenario slices? And does the schema reveal failures that would be invisible if evaluation were limited to coarse source metadata or free-form descriptions? Those are the benchmark questions CARScenes is intended to support.

7 Conclusion

We reframed CARScenes as an annotation-only dataset and benchmark for structured scene understanding and failure analysis in autonomous driving rather than as a control-oriented or VLA resource. The current canonical release contains 5,192 validated records, a stable schema, fixed split manifests, reusable slice definitions, Croissant metadata, and executable release QA tooling. Its scientific role is to make scene-level failures easy to measure, query, and compare across Argoverse1, Cityscapes, KITTI, and nuScenes.

The released benchmark is built from a VLM-assisted workflow followed by full human correction, so the final CARScenes records should be interpreted as human-reviewed ground truth rather than as raw model output. The next scientific step is therefore broader benchmark coverage rather than basic label verification: more baselines, more slice-wise reporting, and stronger comparative evidence that CARScenes reveals failures hidden by coarse source metadata alone.

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